

Technical Information **KORLOY**

Railway Industry

Railway Structural Components and Applicable Tools / Application Example / Product Details

RAILWAY INDUSTRY

- 01 _ Rail Repair
- 02 _ Axle
- 03 _ Wheel
- 04 _ Rail & Crossing
- 05 _ Wheel Repair
- 06 _ Bogie



Rail Repair



Axle



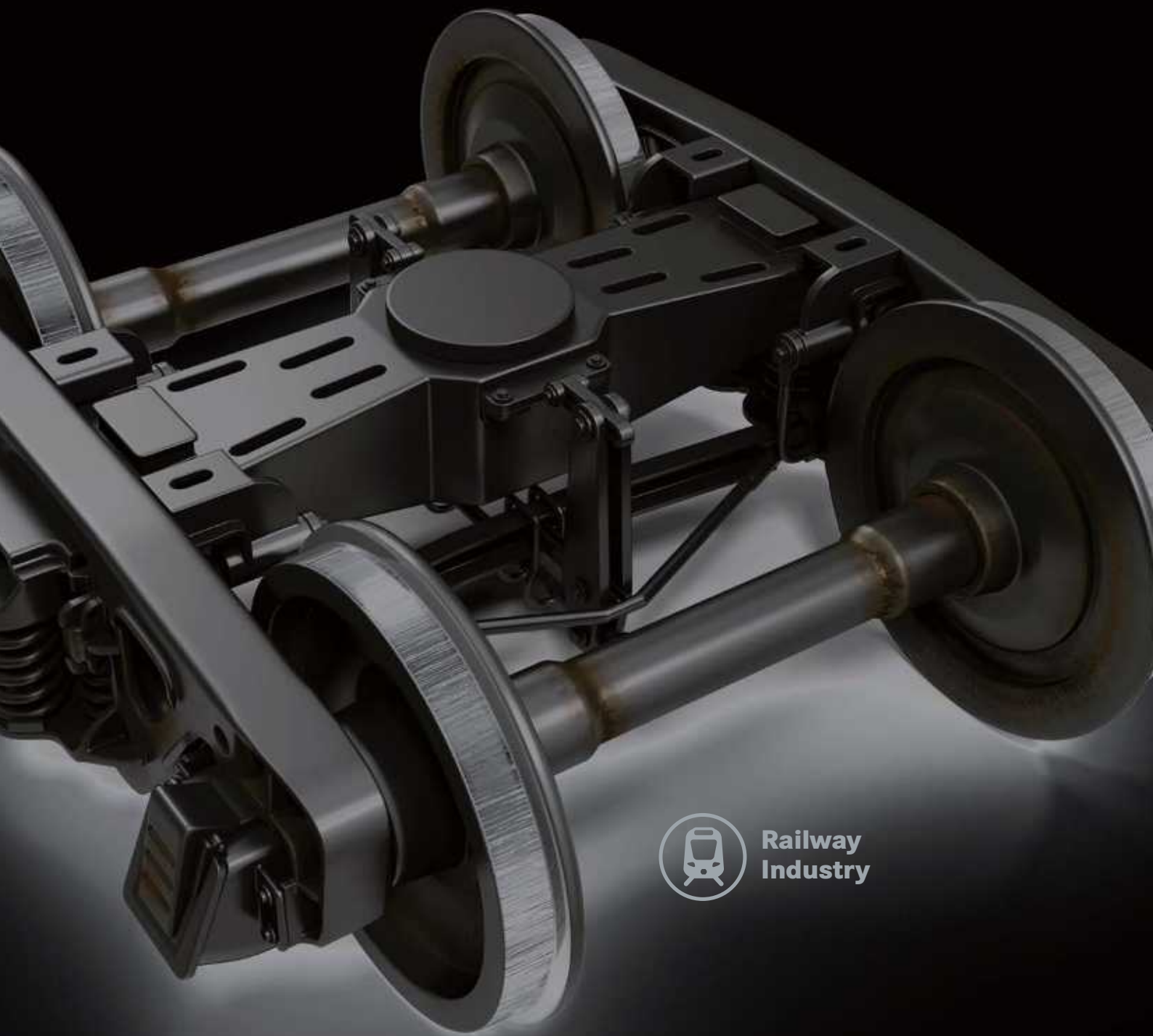
Wheel



Wheel Repair



Bogie



**Railway
Industry**

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6	Bogie	018

Part 2 Application Example

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2	Application Example for Axle	021
3	Application Example for Wheel	023
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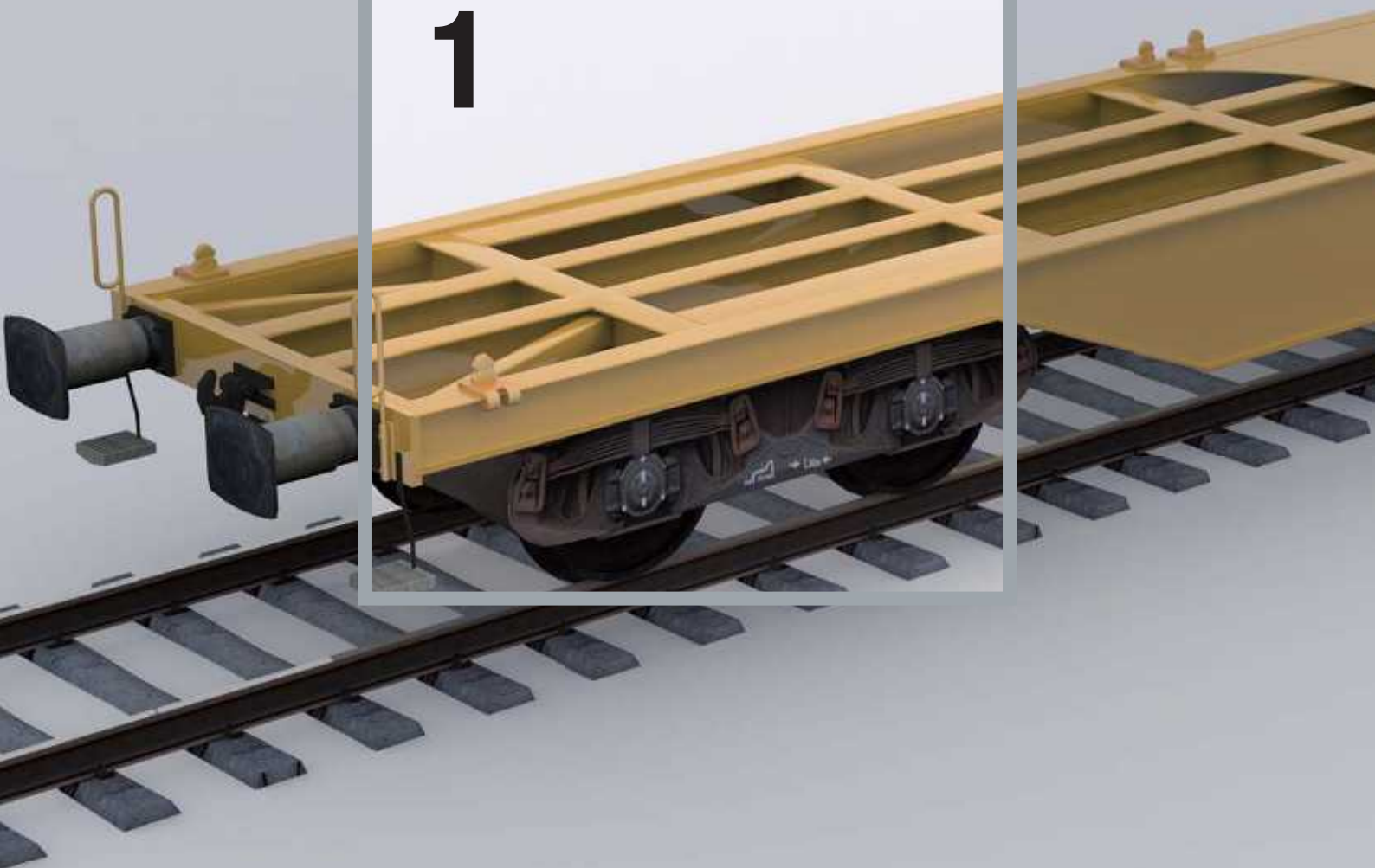
Part 3 Product Details

1	Turning	028
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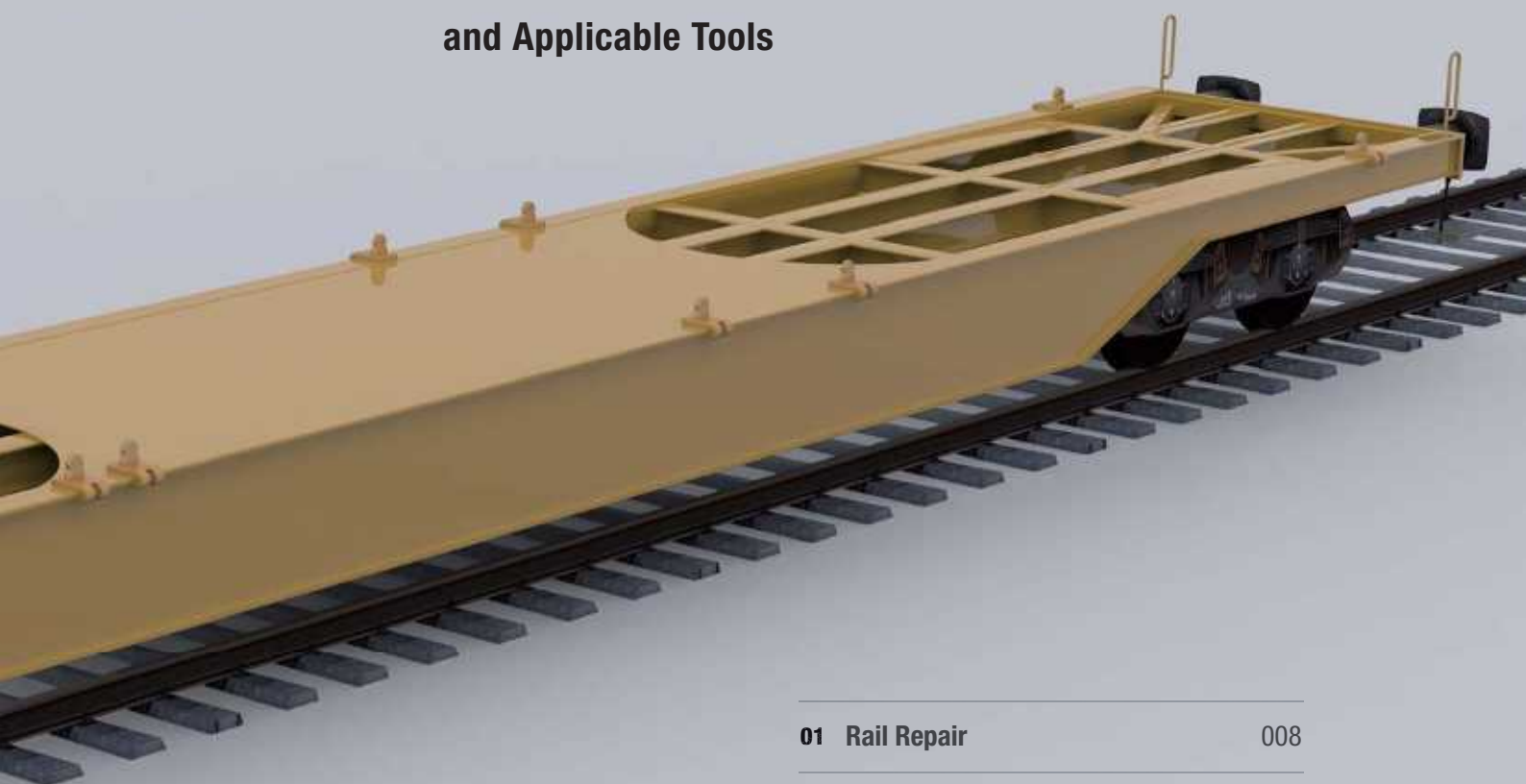


RAILWAY INDUSTRY

Part 1



Railway Structural Components and Applicable Tools



01	Rail Repair	008
02	Axle	010
03	Wheel	012
04	Rail & Crossing	014
05	Wheel Repair	016
06	Bogie	018

Rail Repair

01 Cutter for Rail Repair

<Wedge type(60KgK-SMRP-RW)>

The number of edges : 128

Cutters for repairing the shape of rail with abnormal wear





02 Cutter for Rail Repair

<Screw on type(60KgK-SMRP-RS01)>

The number of edges : 198

Cutters for repairing the shape of rail with abnormal wear

Axle

01 Inserts for Heavy Duty Machining

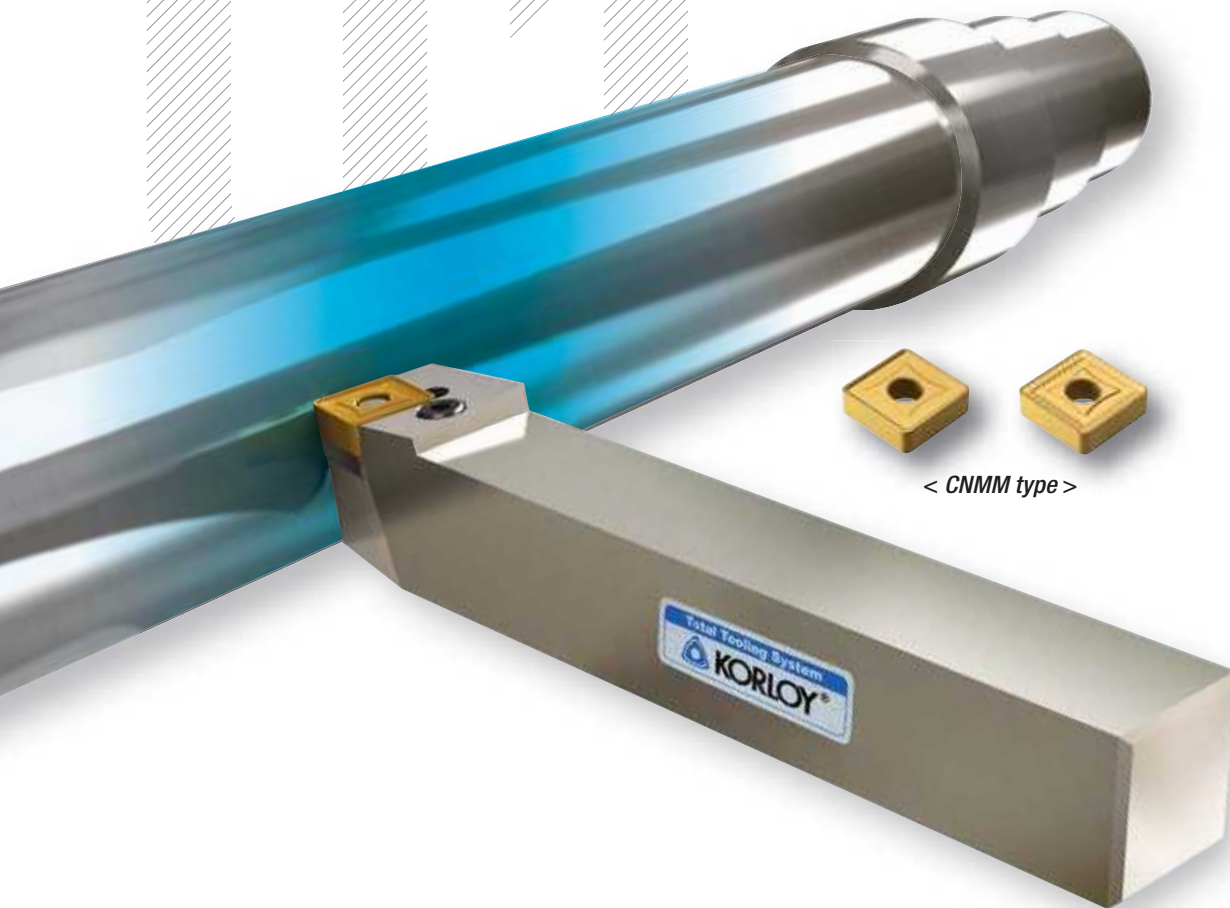
<CNMM>

[External / Profile]

Turning for heavy duty machining

➡ Application Example: P.21~23

➡ Product Details: P.28



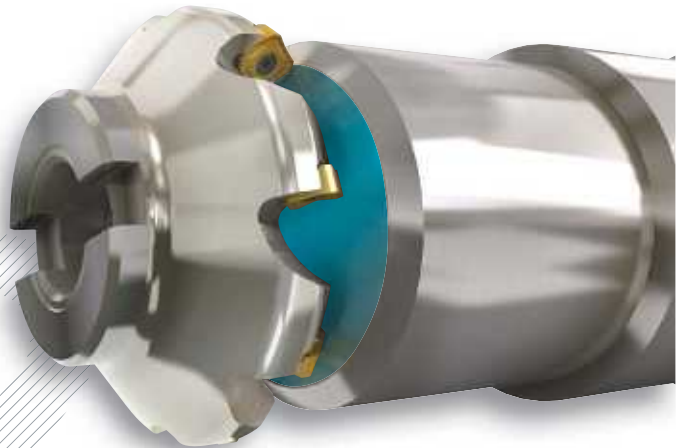
02 Rich Mill

<RM8>

[External / Profile]

Roughing and finishing for top surface facing

➡ Product Details: P.32





03 Holders

<PRDCN(RCMX)>

[External / Profile]

Turning for heavy duty machining

Application Example: P.21~23

Product Details: P.29

04 TPDC Plus Drill

[Hole Making]

Application Example: P.23

Product Details: P.35~36

05 Inserts for Heavy Duty Machining

<SNMM>

[External / Profile]

Turning for heavy duty machining

Application Example: P.21~23

Product Details: P.28



< SNMM type >



Wheel

01 LNUX Insert for Rail Way Wheel

<LNUX>

[External / Profile-Finishing]

➡ Application Example: P.23~24

➡ Product Details: P.29



< LNUX type >



02 Holders

<PRDCN(RCMX)>

[Wheel External / Profile]

Turning for heavy duty machining

➡ Application Example: P.23~24

➡ Product Details: P.29



< RCMX type >





03 Inserts for Heavy Duty Machining

<SNMM/CNMM>

[External / Profile]

Turning for heavy duty machining

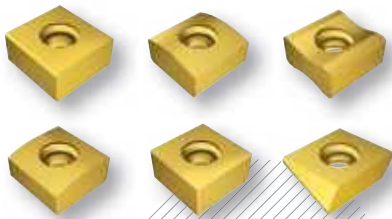
➡ Application Example: P.23~24

➡ Product Details: P.28

Rail & Crossing

01 Forming Cutter for the Top Side of Rail Way

- Cutter diameter: $\varnothing 300(\text{mm})$
- Number of tooth: 33
- One body design of cutter and arbor provides high rigidity



02 Periphery Face Milling for the Top Side of Rail Way

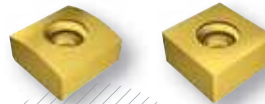
- Cutter diameter: $\varnothing 240(\text{mm})$
- Number of tooth: 25





03 Forming Cutter for the Top Side of Rail Way

- Cutter diameter: Ø160(mm)
- Number of tooth: 16
- Precise forming of rail way is possible



04 Rail Separator Joint Face Milling Cutter

- Cutter diameter: Ø160(mm)
- Number of tooth: 54
- Special customizing is available upon customer's requests



Wheel Repair

01 Inserts for Heavy Duty Machining

<SNMM/CNMM>

[External / Profile]

Turning for heavy duty machining

⇒ Product Details: P.28



< SNMM/CNMM type >



02 LNUX Insert for Wheel Repair

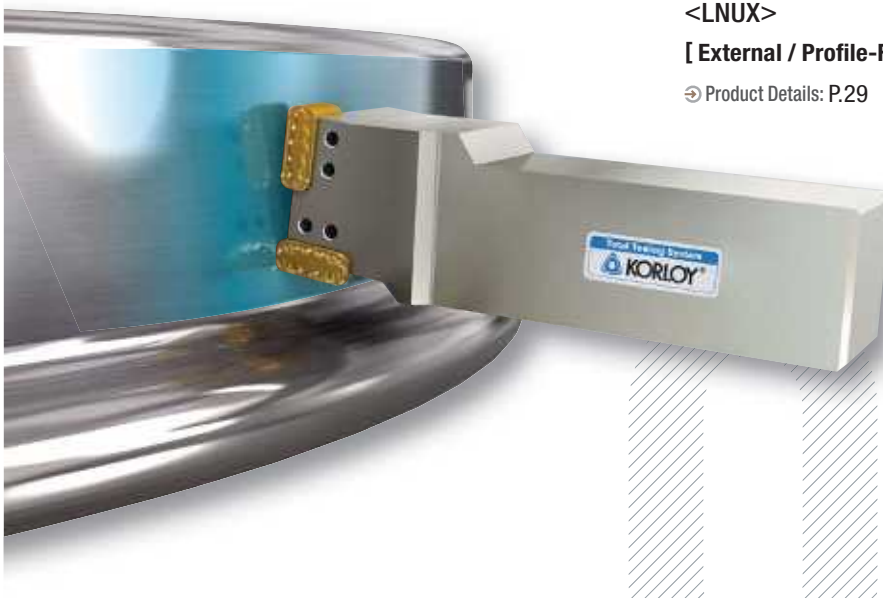
<LNUX>

[External / Profile-Finishing]

⇒ Product Details: P.29



< LNUX type >





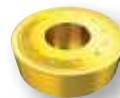
03 Holders

<PRDCN(RCMX)>

[External / Profile]

Turning for heavy duty machining

➔ Product Details: P.29



< RCMX type >

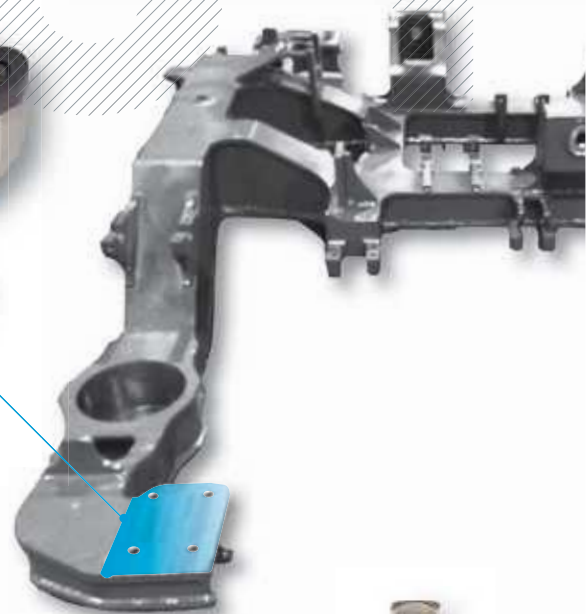
Bogie

01 FMR

[Facing]

➡ Application Example: P.26

➡ Product Details: P.32



02 King Drill

[Hole Making]

➡ Product Details: P.33~34



03 Tap Series

[Tapping]

➡ Application Example: P.25~26





RAILWAY INDUSTRY

Part 2

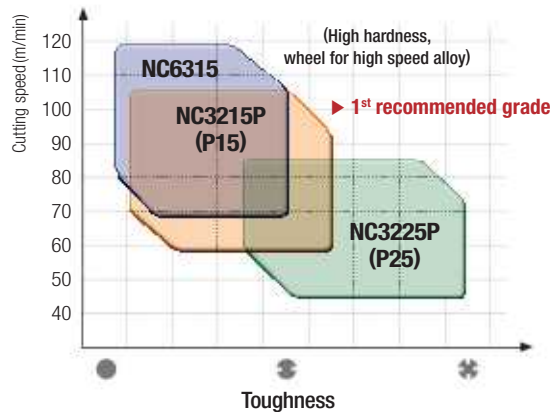
Application Example

01	Grade & Chip Breaker	020
02	Application Example for Axle	021
03	Application Example for Wheel	023
04	Application Example for Bogie	022

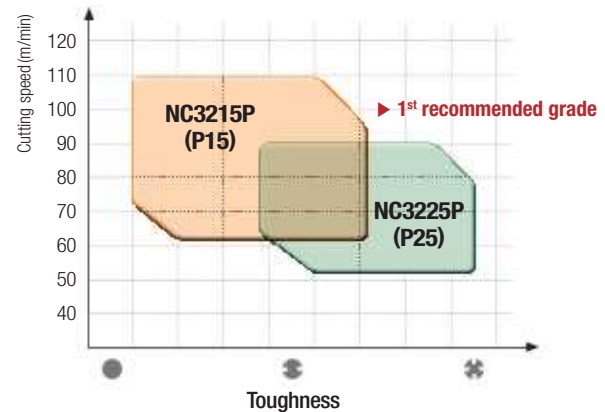
Grade & Chip Breaker

Line-up for grades of new and repaired railway industry wheel

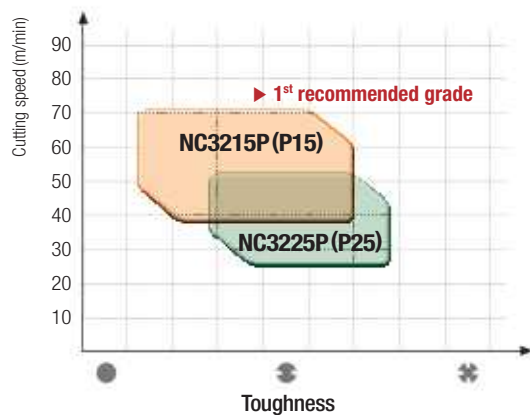
New grade line-up for wheel



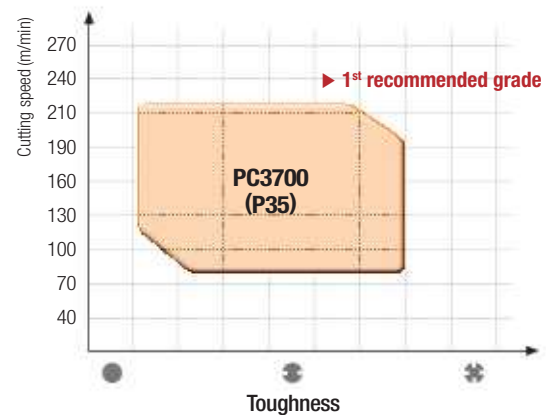
Grade line-up for axle



Grade line-up for repaired wheel

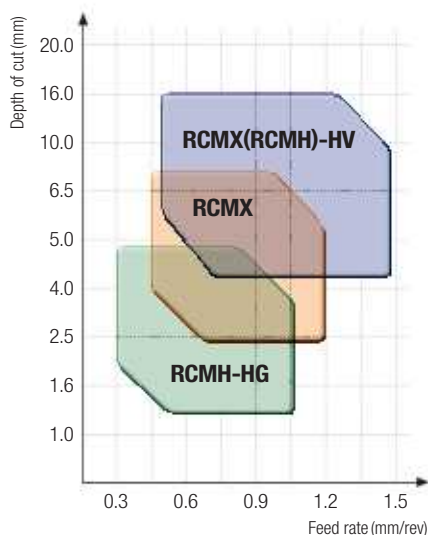


Grade line-up for railway machining

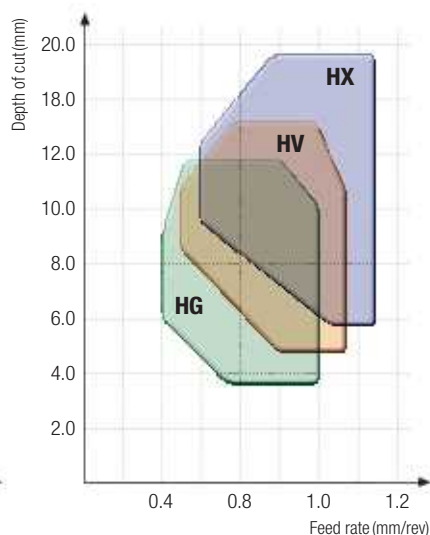


Chip breaker line-up for new and repaired railway wheel

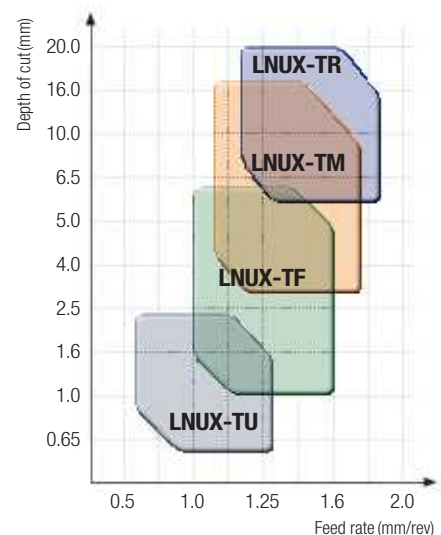
New wheel machining/ R type (25, 32)



New wheel and axle machining/ C, SNMM (19, 25)



Repaired wheel LNUX (19, 30)





Axle

Application Example 1

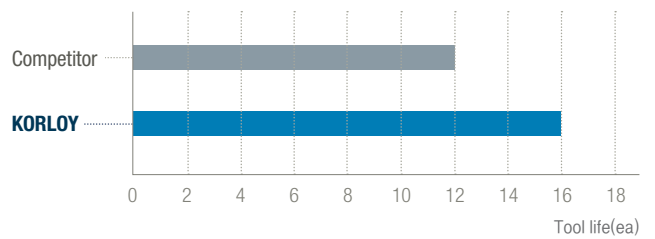


Cutting conditions

Insert	SNMM250724-HV
Grade	NC3225P
Workpiece	45C
Cutting speed	$vc = 160 \sim 180$ (m/min)
Feed	$fn = 0.7 \sim 1.1$ (mm/rev)
Depth of cut	$ap = 5 \sim 20$ (mm)
Coolant	Wet

Maker	KORLOY	Competitor
SNMM250724	16ea	12ea

[Tool life]



Application Example 2

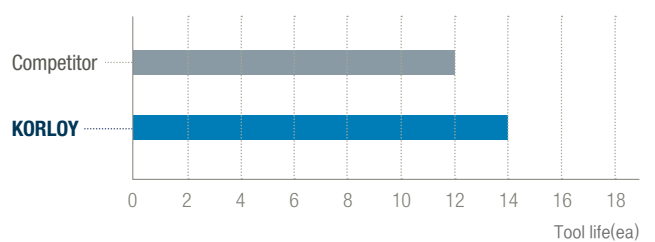


Cutting conditions

Insert	CNMG120408-CP
Grade	NC3215P
Workpiece	45C
Cutting speed	$vc = 230$ (m/min)
Feed	$fn = 0.27$ (mm/rev)
Depth of cut	$ap = 1.5$ (mm)
Coolant	Wet

Maker	KORLOY	Competitor
CNMG120408	14ea	12ea

[Tool life]



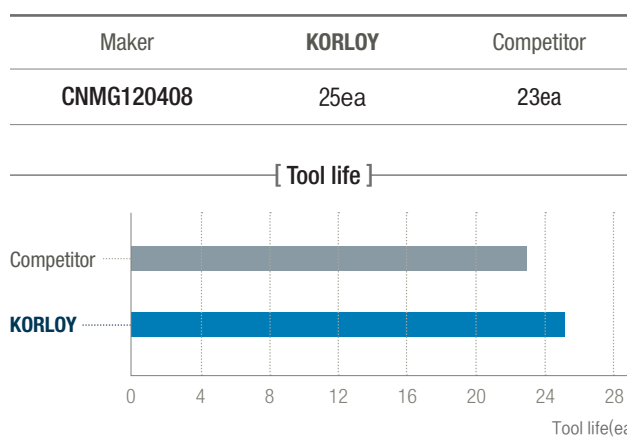
Axle

Application Example 3

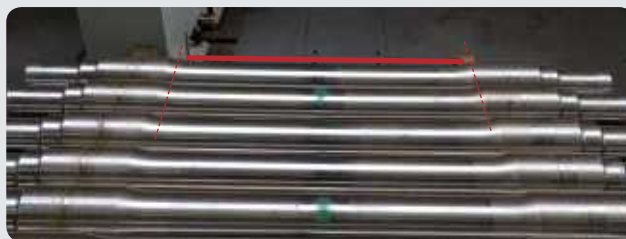


Cutting conditions

Insert	CNMG120408-CP
Grade	NC3215P
Workpiece	45C
Cutting speed	$vc = 84$ (m/min)
Feed	$fn = 0.6$ (mm/rev)
Depth of cut	$ap = 4.4$ (mm)
Coolant	Wet

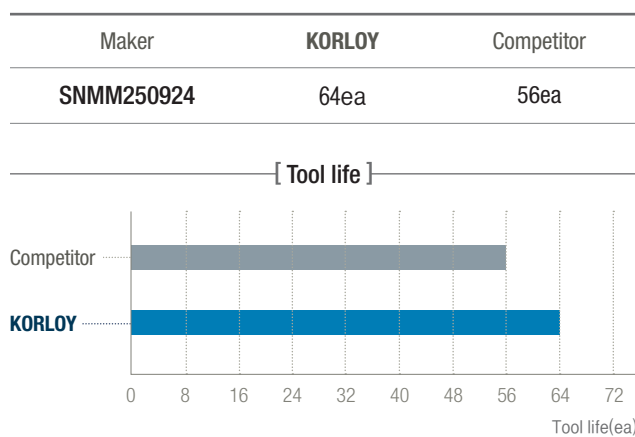


Application Example 4



Cutting conditions

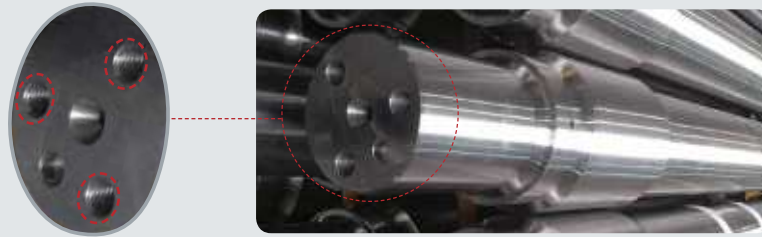
Insert	SNMM250924-HG
Grade	NC3215P
Workpiece	45C
Cutting speed	$vc = 176$ (m/min)
Feed	$fn = 0.2$ (mm/rev)
Depth of cut	$ap = 8 \sim 12$ (mm)
Coolant	Wet





Axle

Application Example 5



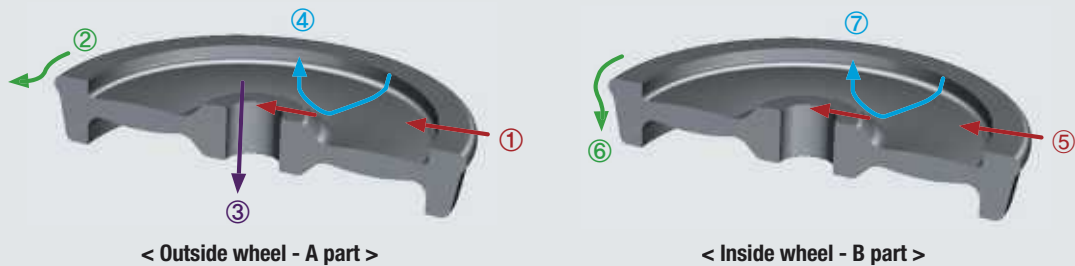
Cutting conditions

Holder	TPDC3D-21025-63
Insert	TPD2100CP
Grade	PC5335
Workpiece	45C
Cutting speed	vc = 62 (m/min)
Feed	fn = 0.3 (mm/rev)
Depth of cut	ap = 12 (mm), Closed hole
Coolant	Wet

Maker	KORLOY	Competitor
Drill(Corn type)	Noise / Vibration decrease → get order	Not good Noise / Vibration

Wheel

Application Example 1



Cutting conditions

Insert	SNMM250724-HX ①/②/④ SNMM250724-HG ③
Grade	NC3215P(1 st), NC6315P(2 nd)
Workpiece	High carbon steel rolling wheel
Cutting speed	vc = 100 (m/min)
Feed	fn = 1.0 ~ 1.5 (mm/rev)
Depth of cut	ap = 2 (mm)
Coolant	Wet

Maker		KORLOY	A Co.	B Co.	C Co.
RCMH3209M0	A Part	19ea	19ea	15ea	19ea
	B Part	19ea	19ea	15ea	19ea
	Normal	19(100%)	19(100%)	15(79%)	19(100%)
SNMM250724	A Part	20(200%)	16(80%)	-	8(80%)

Wheel

Application Example 2



Cutting conditions

Insert	SNMM250724-HV
Grade	NC3215P(1 st), NC6315P(2 nd)
Workpiece	High carbon steel rolling wheel
Cutting speed	$vc = 150$ (m/min)
Feed	$fn = 1.1$ (mm/rev)
Depth of cut	$ap = 6 \sim 7$ (mm)
Coolant	Air

Maker	KORLOY	Competitor
SNMM250724	Excellent chip evacuation	Not good chip evacuation

Application Example 3

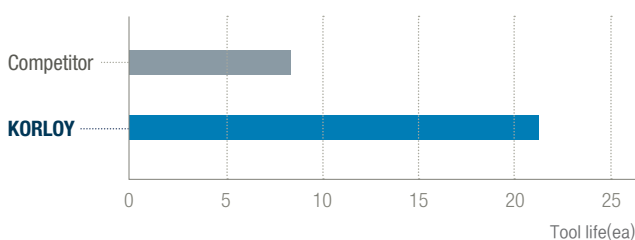


Cutting conditions

Insert	RCMX2507M0
Grade	NC3215P(1 st), NC3225P(2 nd)
Workpiece	Carbon Steel
Cutting speed	$vc = 90$ (m/min)
Feed	$fn = 1.2 \sim 1.4$ (mm/rev)
Depth of cut	$ap = 3 \sim 8$ (mm)
Coolant	Wet

Maker	KORLOY	Competitor
RCMX2507M0	21ea	8ea

[Tool life]





Bogie

Application Example 1

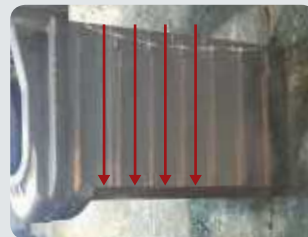


Cutting conditions

Holder	AMCM4085HM-5Z76-CH	
Insert	Internal	APMT180650R-MM
	External	APMT1806PDSR-MM
Grade	PC3545	
Workpiece	Beam Bolster(SCMn1 - Cast Steel)	
Cutting speed	vc = 158 (m/min)	
Feed	fz = 0.37 (mm/t)	
No. of tooth	25(5F*5T)	
Depth of cut	ap = 40 (mm), ae = 3 (mm)	
Coolant	Wet	

Maker	KORLOY	Competitor
Milling Multi type (Good machinability)	Noise / Vibration decrease → get order	Not good Noise / Vibration

Application Example 2



Cutting conditions

Holder	AMCM4080HM-5Z61-CH
Insert	APMT1806PDSR-MM
Grade	PC3545
Workpiece	Front Support(SCMn1 - Cast Steel)
Cutting speed	vc = 158 (m/min)
Feed	fz = 0.37 (mm/t)
No. of tooth	20(5F*4T)
Depth of cut	ap = 40 (mm), ae = 3 (mm)
Coolant	Wet

Maker	KORLOY	Competitor
Milling Multi type (Chip removal amount)	56,264cm ³ 28ea	50,200cm ³ 25ea

Bogie

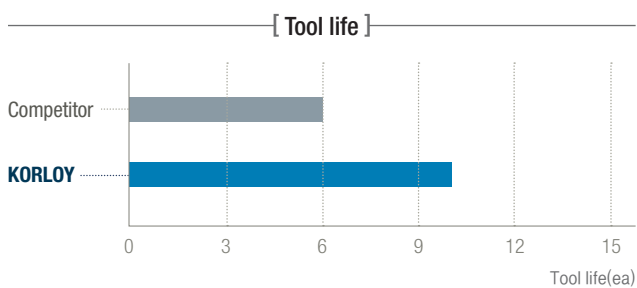
Application Example 3



Cutting conditions

Holder	FMRCM6080HRX-5-CH
Insert	RPMX2007M0S-MM
Grade	NCM535(1 st), NCM545(2 nd)
Workpiece	Front Support(SCMn1 - Cast Steel)
Cutting speed	$vc = 160.8 \text{ (m/min)}$
Feed	$fz = 0.63 \text{ (mm/t)}$
No. of tooth	5
Depth of cut	$ap = 3 \text{ (mm)}, ae = 0.8D$
Coolant	Wet

Maker	KORLOY	Competitor
Round (Chip removal amount)	13,650cm ³ 10ea	8,370cm ³ 6ea





RAILWAY INDUSTRY

Part 3

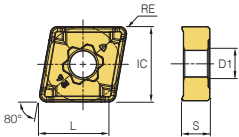
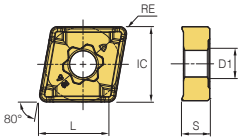
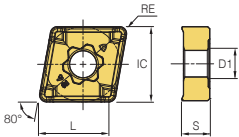
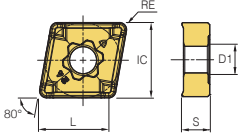
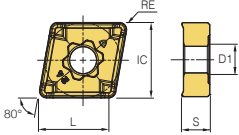
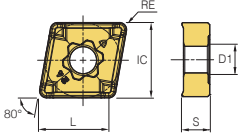
Product Details

01	Turning	028
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02	Milling	030
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03	Drilling	033
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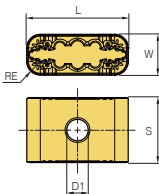
Turning

Picture	Designation	Coated			Dimensions(mm)					Cutting condition		Geometries
		NC6315	NC3215P	NC3225P	L	IC	S	RE	D1	fn (mm/rev)	ap (mm)	
	CNMM 190612-HG				19.344	19.05	6.35	1.2	7.93	0.30~0.90	3.0~10.0	
	190616-HG				19.344	19.05	6.35	1.6	7.93	0.35~0.95	3.0~10.0	
	190624-HG				19.344	19.05	6.35	2.4	7.93	0.40~1.00	3.0~10.0	
	250924-HG				25.792	25.4	7.94	2.4	9.12	0.40~1.20	3.5~13.0	
	250932-HG				25.792	25.4	7.94	3.2	9.12	0.50~1.20	3.5~13.0	
	CNMM 190616-HV				19.344	19.05	6.35	1.6	7.93	0.45~1.05	4.0~11.0	
	190624-HV				19.344	19.05	6.35	2.4	7.93	0.50~1.10	4.0~11.0	
	250924-HV				25.792	25.4	7.94	2.4	9.12	0.50~1.40	4.0~15.0	
	CNMM 190616-HX				19.344	19.05	6.35	1.6	7.93	0.55~1.10	4.5~15.0	
	190624-HX				19.344	19.05	6.35	2.4	7.93	0.60~1.20	4.5~15.0	
	250924-HX				25.792	25.4	7.94	2.4	9.12	0.60~1.50	4.5~18.0	
	SNMM 190612-HG				19.05	19.05	6.35	1.2	7.93	0.30~0.90	3.0~10.0	
	190616-HG				19.05	19.05	6.35	1.6	7.93	0.35~0.95	3.0~10.0	
	190624-HG				19.05	19.05	6.35	2.4	7.93	0.40~1.00	3.0~10.0	
	250924-HG				25.4	25.4	7.94	2.4	9.12	0.40~1.20	3.5~13.0	
	250932-HG				25.4	25.4	7.94	3.2	9.12	0.50~1.20	3.5~13.0	
	SNMM 190616-HV				19.05	19.05	6.35	1.6	7.93	0.45~1.05	4.0~11.0	
	190624-HV				19.05	19.05	6.35	2.4	7.93	0.50~1.10	4.0~11.0	
	250724-HV				25.4	25.4	6.35	2.4	9.12	0.50~1.40	4.0~15.0	
	250924-HV				25.4	25.4	7.94	2.4	9.12	0.50~1.40	4.0~15.0	
	SNMM 190616-HX				19.05	19.05	6.35	1.6	7.93	0.55~1.10	4.5~15.0	
	190624-HX				19.05	19.05	6.35	2.4	7.93	0.60~1.20	4.5~15.0	
	250924-HX				25.4	25.4	7.94	2.4	9.12	0.60~1.50	4.5~18.0	

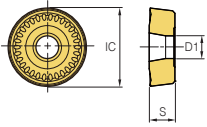
●: Stock item



Turning

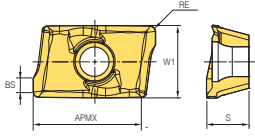
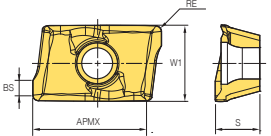
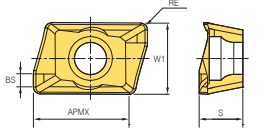
Picture	Designation	Coated			Dimensions (mm)					Cutting condition		Geometries
		NC6315	NC3215P	NC3225P	L	W	S	RE	D1	fn (mm/rev)	ap (mm)	
	LNUX 191940-TU				19.0	19.0	19.2	4.0	6.35	0.5~1.0	1.0~8.0	
	301940-TU				30.0	19.0	19.2	4.0	6.35	0.5~1.0	1.0~18.0	
	LNUX 191940-TF				19.0	19.0	19.2	4.0	6.35	0.6~1.1	1.5~9.0	
	301940-TF				30.0	19.0	19.2	4.0	6.35	0.6~1.1	1.5~19.0	
	LNUX 191940-TM				19.0	19.0	19.2	4.0	6.35	0.7~1.2	2.0~10.0	
	301940-TM				30.0	19.0	19.2	4.0	6.35	0.7~1.2	2.5~20.0	
	LNUX 191940-TR				19.0	19.0	19.2	4.0	6.35	0.8~1.3	2.0~12.0	
	301940-TR				30.0	19.0	19.2	4.0	6.35	0.8~1.3	2.5~22.0	

●: Stock item

Picture	Designation	Coated			Dimensions (mm)					Cutting condition		Geometries
		NC6315	NC3215P	NC3225P	L	IC	S	RE	D1	fn (mm/rev)	ap (mm)	
	RCMX 2507M0				-	25.0	7.94	-	7.25	0.15~1.15	4.0~12.0	
	3209M0				-	32.0	9.52	-	9.55	0.6~1.45	5.0~14.5	
	RCMX 2507M0-HV				-	25.0	7.94	-	7.25	0.55~1.2	4.5~12.5	
	3209M0-HV				-	32.0	9.52	-	9.55	0.65~1.5	5.5~15.5	
	RCMH 2507M0-HG				-	25.0	7.94	-	7.5	0.45~1.1	3.5~11.5	
	3209M0-HG				-	32.0	9.52	-	10.3	0.55~1.4	4.5~14.0	
	RCMH 2507M0-HV				-	25.0	7.94	-	7.5	0.55~1.2	4.5~12.5	
	3209M0-HV				-	32.0	9.52	-	10.3	0.65~1.5	5.5~15.5	

●: Stock item

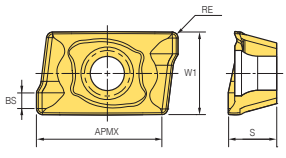
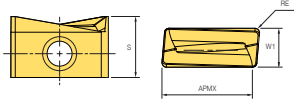
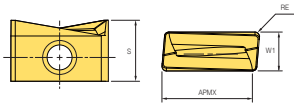
Milling

Inserts	Designation		Coated			Dimensions(mm)					Geometries
			PC3545	PC5300	PC5400	W1	S	RE	BS	APMX	
APMT-MF 	APMT	11T3PDSR-MF		●		6.467	3.6	0.5	1.2	11.2	
		1604PDSR-MF		●		9.41	5.76	0.8	1.1	16.4	
		1806PDSR-MF		●		10.98	6.35	0.8	2.2	17.4	
		180612PDSR-MF		●		10.98	6.35	1.2	1.8	17.4	
APMT-ML 	APMT	0903PDER-ML		●		6.21	3.6	0.4	-	9.4	
		090308PDER-ML		●		6.21	3.6	0.8	-	9.4	
		11T3PDER-ML		●		6.467	3.6	0.5	-	11.2	
		11T308PDER-ML		●		6.467	3.6	0.8	-	11.2	
		160404PDER-ML		●		9.41	5.76	0.4	-	16.4	
		1604PDER-ML		●		9.41	5.76	0.8	-	16.4	
		180604PDER-ML		●		10.98	6.35	0.4	-	17.4	
		1806PDER-ML		●		10.98	6.35	0.8	-	17.4	
		180612PDER-ML		●		10.98	6.35	1.2	-	17.4	
		180616PDER-ML		●		10.98	6.35	1.6	-	17.4	
		180620PDER-ML		●		10.98	6.35	2.0	-	17.4	
		180624PDER-ML		●		10.98	6.35	2.4	-	17.4	
		180630R-ML		●		10.98	6.35	3.0	-	17.4	
APMT-MM 	APMT	060202PDSR-MM		●		4.24	2.6	0.2	1.1	6.0	
		0602PDSR-MM		●		4.24	2.6	0.4	0.9	6.0	
		060208PDSR-MM		●		4.24	2.6	0.8	0.5	6.0	
		060212R-MM		●		4.24	2.6	1.2	0.1	6.0	
		060216R-MM		●		4.24	2.6	1.6	-	6.0	
		0903PDSR-MM		●		6.21	3.6	0.4	0.9	9.4	
		090308PDSR-MM		●		6.21	3.6	0.8	0.5	9.4	
		090312R-MM		●		6.21	3.6	1.2	-	9.4	
		090316R-MM		●		6.21	3.6	1.6	-	9.4	
		090320R-MM		●		6.21	3.6	2.0	-	9.2	
		090332R-MM		●		6.21	3.6	3.2	-	9.2	
		11T3PDSR-MM		●		6.467	3.6	0.5	1.2	11.2	
		11T308PDSR-MM		●		6.467	3.6	0.8	0.9	11.2	
		11T312PDSR-MM		●		6.467	3.6	1.2	0.5	11.2	
		11T316R-MM		●		6.467	3.6	1.6	-	11.0	
		11T318R-MM		●		6.467	3.6	1.8	-	11.0	
		11T324R-MM		●		6.467	3.6	2.4	-	11.0	

●: Stock item

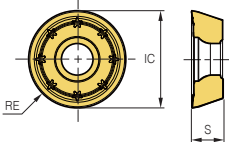
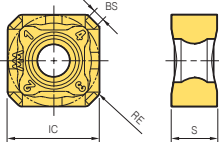
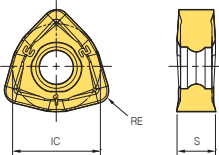
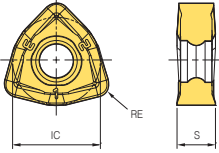
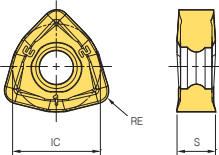


Milling

Inserts	Designation	Coated			Dimensions(mm)					Geometries
		PC3545	PC5300	PC5400	W1	S	RE	BS	APMX	
APMT-MM 	APMT	160404PDSR-MM	●		9.41	5.76	0.4	1.1	16.4	
		1604PDSR-MM	●		9.41	5.76	0.8	1.1	16.4	
		160410PDSR-MM	●		9.41	5.76	1.0	0.9	16.4	
		160416PDSR-MM	●		9.41	5.76	1.6	0.3	16.4	
		160420R-MM			9.41	5.76	2.0	4.5	16.4	
		160424R-MM	●		9.41	5.76	2.4	-	16.0	
		160430R-MM	●		9.41	5.76	3.0	-	16.0	
		160432R-MM	●		9.41	5.76	3.2	-	16.0	
		160450R-MM	●		9.41	5.76	5.0	-	16.0	
		160464R-MM	●		9.41	5.76	6.4	-	16.0	
		1806PDSR-MM	●		10.98	6.35	0.8	2.2	18.7	
		180612PDSR-MM	●		10.98	6.35	1.2	1.8	17.4	
		180616PDSR-MM	●		10.98	6.35	1.6	1.4	17.4	
		180620PDSR-MM			10.98	6.35	2.0	1.0	17.4	
		180624PDSR-MM	●		10.98	6.35	2.4	0.6	17.4	
		180630R-MM	●		10.98	6.35	3.0	-	16.7	
		180632R-MM	●		10.98	6.35	3.2	-	16.7	
		180640R-MM	●		10.98	6.35	4.0	-	16.7	
		180648R-MM	●		10.98	6.35	4.8	-	16.7	
		180650R-MM	●		10.98	6.35	5.0	-	16.7	
		180660R-MM			10.98	6.35	6.0	-	16.7	
		180664R-MM	●		10.98	6.35	6.4	-	16.7	
LNKT-ML 	LNKT	080404PNR-ML			-	6.6	0.4	1.117	8.0	
		080408PNR-ML			-	6.6	0.8	0.717	8.0	
		080412PNR-ML			-	6.6	1.2		8.0	
		080416PNR-ML			-	6.6	1.6		8.0	
		140604PNR-ML			-	10.0	0.4		12.7	
		140608PNR-ML			-	10.0	0.8	1.1273	12.7	
		140612PNR-ML			-	10.0	1.2		12.7	
		140616PNR-ML			-	10.0	1.6		12.7	
		170704PNR-ML			-	11.0	0.4	1.332	16.5	
		170708PNR-ML			-	11.0	0.8	0.932	16.5	
		170712PNR-ML			-	11.0	1.2	0.532	16.5	
		170716PNR-ML			-	11.0	1.6	0.132	16.5	
		170720PNR-ML			-	11.0	2.0	0	16.5	
LNKT-MM 	LNKT	080404PNR-MM			-	6.6	0.4	1.117	8.0	
		080408PNR-MM			-	6.6	0.8	0.717	8.0	
		080412PNR-MM			-	6.6	1.2		8.0	
		080416PNR-MM			-	6.6	1.6		8.0	
		140604PNR-MM			-	10.0	0.4		12.7	
		140608PNR-MM			-	10.0	0.8	1.1273	12.7	
		140612PNR-MM			-	10.0	1.2		12.7	
		140616PNR-MM			-	10.0	1.6		12.7	
		170704PNR-MM			-	11.0	0.4	1.332	16.5	
		170708PNR-MM			-	11.0	0.8	0.932	16.5	
		170712PNR-MM			-	11.0	1.2	0.532	16.5	
		170716PNR-MM			-	11.0	1.6	0.132	16.5	
		170720PNR-MM			-	11.0	2.0		16.5	

● : Stock item

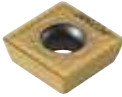
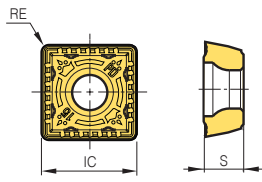
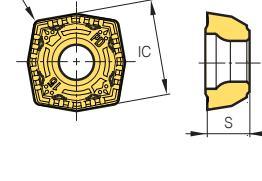
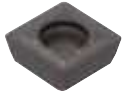
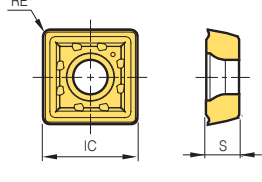
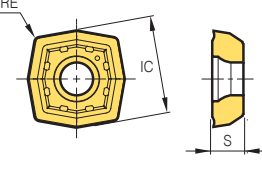
Milling

Inserts	Designation		Coated			Dimensions(mm)					Geometries
			PC3545	PC5300	PC5400	IC	S	RE	BS	APMX	
RPMX-MM 	RPMX	1605M0S-MM				16.0	5.6	-	-	-	
		2006M0S-MM				20.0	6.4	-	-	-	
SNM(E)X-MM 	SNMX	1206ANN-MM				12.7	6.35	0.8	1.56	6.0	
		1507ANN-MM				15.875	7.94	0.8	2.36	7.5	
	SNEX	1206ANN-MM				12.7	6.35	0.8	1.56	6.0	
		1507ANN-MM				15.875	7.94	0.8	2.36	7.5	
WNMX-MF 	WNMX	060312ZNN-MF		●		6.0	3.18	1.2	-	-	
		09T316ZNN-MF		●		9.0	3.97	1.6	-	-	
		130520ZNN-MF		●		13.0	5.56	2.0	-	-	
		160720ZNN-MF		●		16.0	7.0	2.0	-	-	
WNMX-ML 	WNMX	060312ZNN-ML		●		6.0	3.18	1.2	-	-	
		09T316ZNN-ML		●		9.0	3.97	1.6	-	-	
		130520ZNN-ML		●		13.0	5.56	2.0	-	-	
		160720ZNN-ML		●		16.0	7.00	2.0	-	-	
WNMX-MM 	WNMX	060312ZNN-MM		●		6.0	3.18	1.2	-	-	
		09T316ZNN-MM		●		9.0	3.97	1.6	-	-	
		130520ZNN-MM		●		13.0	5.56	2.0	-	-	
		160720ZNN-MM		●		16.0	7.00	2.0	-	-	

●: Stock item

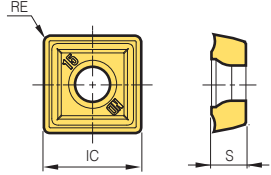
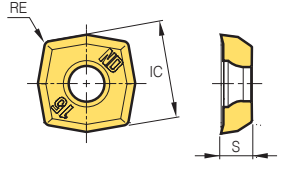


Drilling

Inserts	Designation	Coated				Uncoated	Dimensions (mm)			Geometries
		PC5335	PC5300	PC3700	PC9540	H01	IC	S	RE	
[Peripheral] SPMT-PD  Universal	SPMT	040204-PD	●	●	●		4.7	2.4	0.4	
		050204-PD	●	●	●		5.1	2.4	0.4	
		060205-PD	●	●	●		6.2	2.5	0.5	
		07T208-PD	●	●	●		7.5	2.8	0.8	
		090308-PD	●	●	●		9.2	3.3	0.8	
		11T308-PD	●	●	●		11.0	4.0	0.8	
		130410-PD	●	●	●		13.0	4.5	1.0	
		15M510-PD	●	●	●		15.2	5.0	1.0	
		180510-PD	●	●	●		18.2	5.5	1.0	
[Central] XOMT-PD  Universal	XOMT	040204-PD	●		●		4.9	2.4	0.4	
		050204-PD	●		●		5.4	2.4	0.4	
		060204-PD	●		●		6.6	2.5	0.4	
		07T205-PD	●		●		7.8	2.8	0.5	
		090305-PD	●		●		9.6	3.3	0.5	
		11T306-PD	●		●		11.4	4.0	0.6	
		130406-PD	●		●		13.6	4.5	0.6	
		15M508-PD	●		●		15.9	5.0	0.8	
		180508-PD	●		●		18.9	5.5	0.8	
[Peripheral] SPMT-LD  Mild steel	SPMT	060205-LD	●				6.2	2.5	0.5	
		07T208-LD	●				7.5	2.8	0.8	
		090308-LD	●				9.2	3.3	0.8	
		11T308-LD	●				11.0	4.0	0.8	
		130410-LD	●				13.0	4.5	1.0	
		15M510-LD	●				15.2	5.0	1.0	
		180510-LD	●				18.2	5.5	1.0	
[Central] XOMT-LD  Mild steel	XOMT	060204-LD	●				6.6	2.5	0.4	
		07T205-LD	●				7.8	2.8	0.5	
		090305-LD	●				9.6	3.3	0.5	
		11T306-LD	●				11.4	4.0	0.6	
		130406-LD	●				13.6	4.5	0.6	
		15M508-LD	●				15.9	5.0	0.8	
		180508-LD	●				18.9	5.5	0.8	

● : Stock item

Drilling

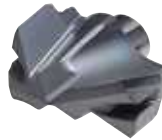
Inserts	Designation		Coated				Uncoated	Dimensions (mm)			Geometries
			PC5335	PC5300	PC3700	PC9540		IC	S	RE	
[Peripheral] SPET-ND  Al	SPET	040204-ND					●	4.7	2.4	0.4	
		050204-ND					●	5.1	2.4	0.4	
		060205-ND					●	6.2	2.5	0.5	
		07T208-ND					●	7.5	2.8	0.8	
		090308-ND					●	9.2	3.3	0.8	
		11T308-ND					●	11.0	4.0	0.8	
		130410-ND					●	13.0	4.5	1.0	
		15M510-ND					●	15.2	5.0	1.0	
		180510-ND					●	18.2	5.5	1.0	
[Central] XOET-ND  Al	XOET	040204-ND					●	4.9	2.4	0.4	
		050204-ND					●	5.4	2.4	0.4	
		060204-ND					●	6.6	2.5	0.4	
		07T205-ND					●	7.8	2.8	0.5	
		090305-ND					●	9.6	3.3	0.5	
		11T306-ND					●	11.4	4.0	0.6	
		130406-ND					●	13.6	4.5	0.6	
		15M508-ND					●	15.9	5.0	0.8	
		180508-ND					●	18.9	5.5	0.8	

● : Stock item

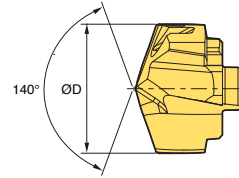
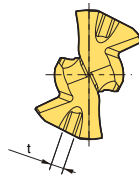


Drilling

TPDB Insert



CP

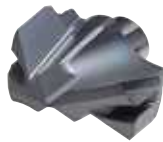


(mm)

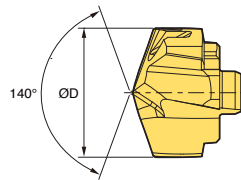
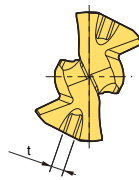
Drill dia. ØD (mm)	P type (CP)	Coated			Holder	Wrench
	TPDC-P	PC5335	PC5300	PC330P		
12.0	TPD1200CP	●			TPDC□D-12016-□	TPDC- W1216
12.2	TPD1220CP	●				
12.5	TPD1250CP	●			TPDC□D-12516-□	
12.6	TPD1260CP	●				
13.0	TPD1300CP	●			TPDC□D-13016-□	
13.5	TPD1350CP	●			TPDC□D-13516-□	
14.0	TPD1400CP	●				
14.2	TPD1420CP	●			TPDC□D-14016-□	
14.3	TPD1430CP	●				
14.5	TPD1450CP	●			TPDC□D-14516-□	
15.0	TPD1500CP	●				
15.5	TPD1550CP	●			TPDC□D-15020-□	
16.0	TPD1600CP	●				
16.3	TPD1630CP	●				
16.5	TPD1650CP	●			TPDC□D-16020-□	
16.7	TPD1670CP	●				
17.0	TPD1700CP	●			TPDC□D-17020-□	TPDC- W1721
17.5	TPD1750CP	●				
17.7	TPD1770CP	●				
18.0	TPD1800CP	●				
18.1	TPD1810CP	●				
18.5	TPD1850CP	●			TPDC□D-18025-□	
18.6	TPD1860CP	●				
18.7	TPD1870CP	●				
19.0	TPD1900CP	●				
19.2	TPD1920CP	●				
19.5	TPD1950CP	●			TPDC□D-19025-□	
19.7	TPD1970CP	●				
20.0	TPD2000CP	●				
20.5	TPD2050CP	●			TPDC□D-20025-□	
21.0	TPD2100CP	●				
21.5	TPD2150CP	●			TPDC□D-21025-□	

Drilling

TPDB Insert



CP



(mm)

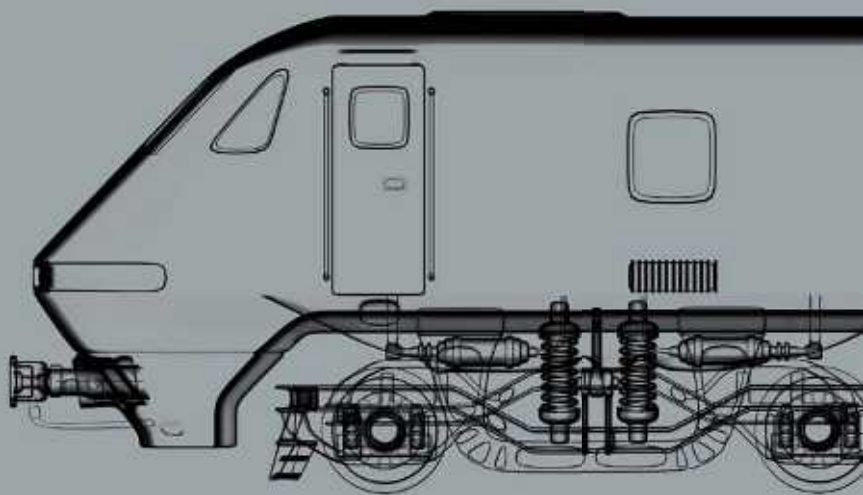
Drill dia. ØD (mm)	P type (CP)	Coated			Holder	Wrench				
	TPDC-P	PC5335	PC5300	PC330P						
22.0	TPD2200CP	●			TPDC□D-22025-□	TPDC- W2225				
22.5	TPD2250CP	●								
22.6	TPD2260CP	●								
22.7	TPD2270CP	●								
23.0	TPD2300CP	●			TPDC□D-23025-□		TPDC- W2225			
23.5	TPD2350CP	●								
24.0	TPD2400CP	●			TPDC□D-24032-□			TPDC- W2225		
24.5	TPD2450CP	●								
25.0	TPD2500CP	●			TPDC□D-25032-□				TPDC- W2225	
25.3	TPD2530CP	●								
25.5	TPD2550CP	●								
25.8	TPD2580CP	●								
25.9	TPD2590CP	●								
26.0	TPD2600CP	●			TPDC□D-26032-□	TPDC- W2630				
26.5	TPD2650CP	●								
27.0	TPD2700CP	●			TPDC□D-27032-□					TPDC- W2630
27.5	TPD2750CP	●								
28.0	TPD2800CP	●			TPDC□D-28032-□		TPDC- W2630			
28.5	TPD2850CP	●								
29.0	TPD2900CP	●			TPDC□D-29032-□			TPDC- W2630		
29.5	TPD2950CP	●								
30.0	TPD3000CP	●			TPDC□D-30032-□				TPDC- W2630	
30.5	TPD3050CP	●								

※ We can provide if you order exact machining specification. Ex) Ø15.9, carbon steel machining → TPDC1590CP/PC337Q ●: Stock item

Parts (applicable wrench)

Picture	Designation		Drill diameter ØD (mm)	Torque (N·m)	Width of clamping part of wrench (mm)
	TPDC-	W1216	12.00 ~ 16.99	2.0 ~ 3.0	1.2
		W1721	17.00 ~ 21.99	2.0 ~ 4.0	1.5
		W2225	22.00 ~ 25.99	3.0 ~ 4.0	2.0
		W2630	26.00 ~ 30.99	4.0 ~ 5.0	2.5

Railway Industry



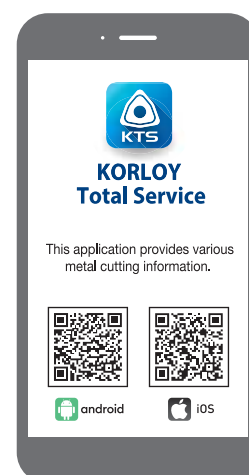
For the safe metalcutting

- Use safety supplies such as protective gloves to prevent possible injury while touching the edge of tools.
- Use safety glasses or safety cover to hedge possible dangers. Inappropriate usage or excessive cutting condition may lead tool's breakage or even the fragment's scattering.
- Clamp the workpiece tightly enough to prevent its movement while its machining.
- Properly manage the tool change phase because the inordinately used tool can be easily broken under the excessive cutting load or severe wear, and it may threat the operator's safety.
- Use safety cover because chips evacuated during cutting are hot and sharp and may cause burns and cuts. To remove chips safely, stop machining, put on protective gloves, and use a hook or other tools.
- Prepare for fire prevention measures as the use of the non-water soluble cutting oil may cause fire.
- Use safety cover and other safety supplies because the spare parts or the inserts can be pulled out due to centrifugal force while high speed machining.



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